ARTICLE IN PRESS

Applied Nursing Research xxx (2014) xxx–xxx

Contents lists available at ScienceDirect

Applied Nursing Research

journal homepage: www.elsevier.com/locate/apnr

The effects of mindfulness eating and yoga exercise on blood sugar levels of pregnant women with gestational diabetes mellitus

Sununta Youngwanichsetha, PhD, MSN *, Sasitorn Phumdoung, PhD, MSN [1], Thitiporn Ingkathawornwong, MSN, BA [2]

Faculty of Nursing, Prince of Songkla University, Hat Yai, Songkhla, Thailand 90112

ARTICLE INFO

Article history:
Received 22 September 2013
Revised 27 January 2014
Accepted 3 February 2014
Available online xxxx

Keywords:
Gestational diabetes
Glycemic control
Pregnant women
Mindfulness eating
Yoga exercise

ABSTRACT

Aim: This randomized controlled trial was carried out to investigate the effect of mindfulness eating and yoga exercise on blood sugar levels among pregnant Thai women with GDM.
Background: Interventions promoting achievement of good glycemic control result in desired pregnancy outcomes. Little is known about the health benefits of mindfulness eating and yoga exercise on blood sugar levels among pregnant with GDM.
Methods: A randomized controlled trial was carried out. Main outcome measures were capillary fasting plasma glucose, 2-h postprandial blood glucose, and hemoglobin A1c.
Results: The intervention group showed significantly reduced fasting plasma glucose, 2-h postprandial blood glucose, and glycosylated hemoglobin (HbA1c) in the intervention group ($p < 0.05$).
Conclusions: Mindfulness eating and yoga exercise had health benefits on glycemic control in pregnant women with GDM. It should be recommended in clinical and community health services.
© 2014 Elsevier Inc. All rights reserved.

1. Introduction

Pregnant women who are overweight or obese and do not take regular exercise regularly are more likely to develop gestational diabetes mellitus (GDM). The prevalence of GDM is increasing worldwide. Between 2 and 9% of pregnant women develop GDM depending on the population. The prevalence of GDM among pregnant women in the United State of America and England was 3–6% and 3.5%, respectively (Bhake & Dayan, 2010). In Thailand, 7.05% of screened pregnant women were diagnosed with GDM (Lueprasitsakul, Teeyapun, Kittivarakul, Srisupundit, & Patumanond, 2008).

GDM is defined as glucose intolerance which is first detected or diagnosed during pregnancy, particular after 24 weeks of gestation (ADA, 2012). Its manifestations include elevated postprandial and fasting blood glucose. According to White's obstetric classification, GDM is classified into two classes: class A1 and class A2. GDM class A1 is diagnosed when fasting plasma glucose is less than 105 mg/dL and 2-h postprandial blood glucose less than 120 mg/dL and can be controlled with medical nutrition therapy alone. GDM class A2 is defined when fasting plasma glucose is more than 105 mg/dL and 2-h postprandial blood glucose is more than 120 mg/dL. It is necessary to lower blood glucose using insulin along with medical nutrition therapy and some kinds of exercise (Gilbert, 2011).

Research evidence reveals that significant factors associated with developing GDM are high pre-pregnancy body mass index, previous GDM, a history of giving birth to large-for-gestational-age baby, and family history of diabetes (Webb, 2013). The odds ratios of developing GDM among women who are overweight, obese, and severely obese are 2.14, 3.56, and 8.56, respectively, when compared with those who are normal weight (Chu, Callaghan, Kim, Schmid, & Lau, 2009).

Pregnant women with GDM are more likely to develop gestational hypertension, preeclampsia, or preterm labor. A serious metabolic complication is diabetic ketoacidosis, which might cause maternal death (Evans, 2009). Maternal hyperglycemia also results in large-for-gestational-age infants, which are at risk for birth injury due to shoulder dystocia. Infants born to mothers with a history of GDM are more likely to develop neonatal hypoglycemia and hyperbilirubinemia (Serlin & Lash, 2009).

Management of GDM aims at lowing fasting or preprandial plasma glucose to less than 95 mg/dl and 2-h postprandial blood glucose to less than 120 mg/dl (ADA, 2012; Ballas, Moore, & Ramos, 2012). Using insulin, medical nutrition therapy and exercise are considered standard guidelines for glycemic control in pregnant women with GDM (ADA, 2012; Webb, 2013). In clinical practice, dietitians and staff nurses educate and counsel these women about carbohydrate choices and low glycemic foods. However, it was reported that around 60% of pregnant women with GDM could not achieve the target of glycemic control (Veeraswamy, Vijayam, Gupta, & Kapur, 2012).

* Corresponding author at: Faculty of Nursing, Prince of Songkla University, Hat Yai, Songkhla, Thailand 90112. Tel.: +66 074 286537; fax: +66 074 286421.
E-mail addresses: sununta.y@psu.ac.th, sununta.y@gmail.com (S. Youngwanichsetha), sasitorn.ph@psu.ac.th (S. Phumdoung), thitiporn.i@psu.ac.th (T. Ingkathawornwong).
[1] Tel.: +66 74 286557; fax: +66 74 286421.
[2] Tel.: +66 74 286547; fax: +66 74 286421.

0897-1897/$ – see front matter © 2014 Elsevier Inc. All rights reserved.
http://dx.doi.org/10.1016/j.apnr.2014.02.002

Please cite this article as: Youngwanichsetha, S., et al., The effects of mindfulness eating and yoga exercise on blood sugar levels of pregnant women with gestational diabetes mellitus, *Applied Nursing Research* (2014), http://dx.doi.org/10.1016/j.apnr.2014.02.002

ARTICLE IN PRESS

Additional and alternative diabetes self-management methods need to be explored. Exercise is suggested for its therapeutic effect of promoting muscular utilization of glucose and improving insulin sensitivity (Jacqueminet & Jannot-Lamotte, 2010). In clinical practice, specific recommendations for pregnant women with GDM are lacking. Moreover, mindfulness eating is widely recommended and practiced for reducing the amount of consumed food and improving eating disorder (Daubenmier et al., 2011). Little is known about the effect of mindfulness eating on glycemic control among pregnant women with GDM. Therefore, this study aimed to determine the effect of mindfulness eating and yoga exercise on blood glucose levels among pregnant Thai women with GDM class A1.

2. Methods

2.1. Research design

A randomized controlled trial was carried out to determine whether mindfulness eating and yoga exercise had an effect on glycemic control of pregnant Thai women with GDM A1. Randomization of participants into the intervention group or the control group was performed by a research assistant using opaque envelopes technique. Participants in the control group were given standard diabetes care while those in the experimental group received standard diabetes care and practiced mindfulness eating in combination with yoga exercise for 8 weeks. Primary outcome measures were capillary fasting and postprandial blood glucose and hemoglobin A1c.

2.2. Participants and procedures

The study was conducted in a tertiary hospital in southern Thailand which is the referral center for diabetes care. Sample size was calculated for two independent samples, intervention group and control group. The primary outcomes of interest were fasting and postprandial blood glucose, and hemoglobin A1c. They were each measured on a continuous scale. Sample size per group was $2c/\text{delta}^2 + 1$, where delta was the standardized effect size[12]. Delta was the value of mean differences of the experimental group and control group divided by the common standard deviation. A two-sided test of 0.05 significant levels was employed. Power for the study was set at 80%. From a pilot study, delta was 0.7. Calculated sample size per group was 85. Then, the total sample size in this study was 170.

Inclusion criteria for recruitment of participants were: 1) pregnant Thai women diagnosed with GDM A1 with 24–30 weeks gestational age, 2) having fasting blood glucose lower than 105 mg/dl, 3) having postprandial blood glucose lower than 120 mg/dl, 4) not receiving insulin therapy for glycemic control, and 5) having no serious complications such as gestational hypertension, preeclampsia, preterm labor, or other serious health problems.

2.3. The intervention

The study protocol and ethical considerations were approved by the IRB of the Faculty of Medicine, Prince of Songkla University, Thailand. The women in the intervention group were trained to perform mindfulness eating and yoga exercise in two 50-minute-sessions. Videos were used in classes, and practicing manuals were offered for all participants to follow. After that they were encouraged to continue mindfulness eating and yoga exercise at home five times a week for 8 weeks. The study protocol comprised two interventions: mindfulness eating and yoga exercise. Mindfulness eating for pregnant women with GDM was developed from the principles of mindful eating (Daubenmier et al., 2011; Solloway & Fisher, 2007) and medical nutrition therapy guidelines (ADA, 2012; Gilbert, 2011). Mindfulness eating for pregnant women with GDM was composed of five steps: 1) setting a goal for blood glucose control, 2) integrating medical nutrition therapy including carbohydrate choices and low glycemic index food, 3) considering portion size, 4) being aware while consuming diabetic food, and 5) eating slowly for 30 to 45 minutes.

Yoga exercise for pregnant women with GDM was developed using yoga pranayama (deep-breathing techniques) and asanas (posture and movements) (Beddoe, Yang, Kenedy, Weiss, & Lee, 2009; Gore, Vaze, Kulkarni, & Oak, 2008; Sahay & Tiwari, 2009). It was designed for 15–20 minutes daily practice for 5 days per week for 8 weeks. Each posture should be repeated for ten times. Yoga asanas for pregnant women comprised 9 postures namely 1) padmasana, 2) brahma mudra, 3) shoulder circling, 4) modified parvatasana, 5) modified gomukhasana, 6) modified tadasana, 7) modified chakrasana, 8) modified bharadvajasana, and 9) dandasana.

3. Data collection

Potential participants who met the study criteria were approached by a registered nurse who had been assigned as a research assistant. After that the researcher informed them about the study objectives and their involvement in the protocol. After obtaining informed consent, baseline data including age, educational level, employment, number of pregnancy, family history of diabetes, previous history of GDM, and diabetic complications were recorded. Practicing mindfulness eating and yoga exercise were encouraged and monitored by the research team every week by phone and on appointment date. Main outcomes measures were fasting, postprandial blood glucose, and hemoglobin A1c. In this study, capillary blood glucose was used for glucometer, and venous blood was sampled for HbA1ctesting. A1c was tested in a hospital laboratory using a method that is certified by the National Glycohemoglobin Standardization Program (NGSP) certified and standardized to the Diabetes Control and Complications Trial (DCCT) assay.

4. Data analysis

Demographic data were analyzed using frequency and percentage and chi square. Continuous data of body weight, body mass index,

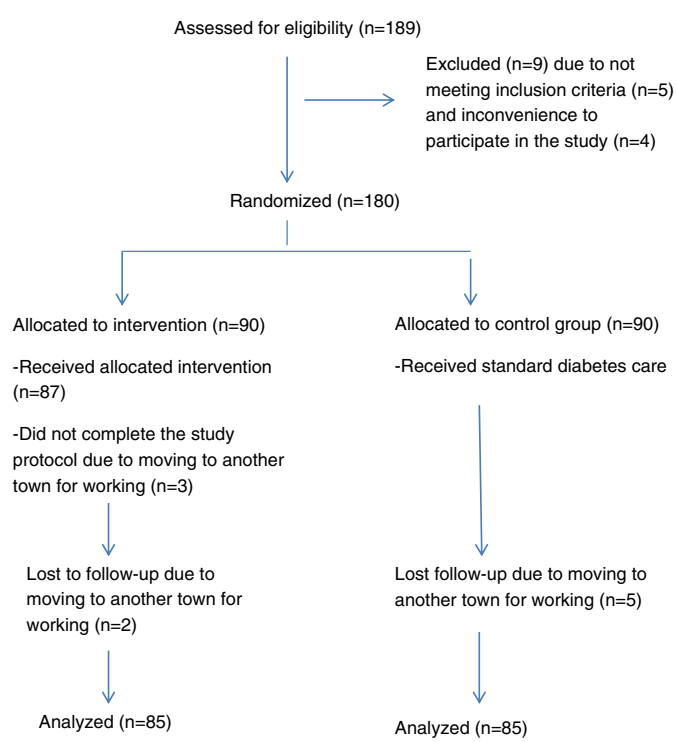

Fig. 1. Flow diagram of the study.

Please cite this article as: Youngwanichsetha, S., et al., The effects of mindfulness eating and yoga exercise on blood sugar levels of pregnant women with gestational diabetes mellitus, *Applied Nursing Research* (2014), http://dx.doi.org/10.1016/j.apnr.2014.02.002

ARTICLE IN PRESS

Table 1
Frequency, percentage, chi square, and *p*-values of participant's characteristics ($n = 170$).

Characteristics	Intervention group ($n = 85$)		Control group ($n = 85$)		Chi square	*p*-values
Religion					0.328	0.536
Buddhist	50	29.42	46	27.05		
Islam	35	20.58	39	22.95		
Education					0.924	0.630
High school	22	12.94	18	10.58		
Vocational degree	35	20.58	32	18.82		
Bachelor degree or higher	28	16.47	35	20.58		
Current employment					0.196	0.657
No	42	24.70	38	22.35		
Yes	43	25.29	47	27.64		
Number of pregnancy					0.659	0.416
1	20	11.76	18	10.58		
2	36	21.17	34	20.00		
3 or higher	29	17.05	33	19.42		
Previous GDM					1.75	0.185
No	57	33.52	52	30.58		
Yes	28	16.47	33	19.42		
Family history of diabetes					0.62	0.431
No	37	21.76	42	24.70		
Yes	48	28.23	43	25.29		

fasting, postprandial blood glucose levels, and hemoglobin A1c were analyzed using mean and standard deviation. The hypothesis testing of the mean differences were performed with independent sample t-test and paired t-test. The significance level was set at 0.05 (two-tailed).

5. Results

A total of 189 pregnant Thai women with GDM A1 were approached and assessed for eligibility. Nine women were excluded because their blood glucose was higher than 120 mg/dL and they received insulin therapy for glycemic control ($n = 5$) or because they were unable to participate as they lived far away from the setting ($n = 4$). The remaining 180 participant were randomly allocated to the intervention group ($n = 90$) and the control group ($n = 90$). All of the participants in the intervention group participated in two sessions of practicing mindfulness eating and yoga exercise in class and completed more than 80% of sessions practiced at home by themselves. No adverse events occurred during practice. The recruitment and retention of participants are summarized in Fig. 1.

The educational background of most participants was vocational level, and over half were employed. Most of them were multiparous, had a family history of diabetes and no serious diabetic complications (Table 1). Mean ages of the pregnant women in the intervention group and control group were 32.58 (SD = 5.01), 31.24 (SD = 4.54) years, respectively. Mean fasting plasma glucose before taking a 100-g oral glucose tolerance test was 89.18 (SD = 12.84), and 89.36 (SD = 13.19), respectively. Baseline characteristics did not differ significantly between groups.

Table 2
Baseline characteristics ($n = 170$).

Characteristics	Intervention ($n = 85$)		Control ($n = 85$)		*p*-values
	Mean	SD	Mean	SD	
Age (year)	32.58	5.01	31.24	4.54	0.347
Body weight (kg)	67.83	10.10	67.40	11.71	0.428
Body mass index (kg/m^2)	27.09	3.56	27.05	4.06	0.525
Fasting plasma glucose (mg/dL)	89.18	8.84	89.36	8.19	0.325
1-h plasma glucose (mg/dL)	205.48	15.28	200.8	15.22	0.346
2-h plasma glucose (mg/dL)	180.85	23.61	176.18	18.78	0.725
3-h plasma glucose (mg/dL)	148.24	20.88	149.85	13.14	0.645

Pregnant Thai women with GDM A1 in the intervention group had significantly lower fasting, postprandial blood glucose, and hemoglobin A1c than those in the control group. Mean fasting blood glucose in the intervention and control group was 83.39 (SD = 17.69) and 85.85 (SD = 17.94) mg/dL) ($p = 0.012$). Mean postprandial blood glucose in the intervention and control group was 105.67 (SD = 12.93) and 112.36 (SD = 13.15) mg/dL ($p = 0.001$). Mean hemoglobin A1c in the intervention and control group was 5.23% (SD = 0.72) and 5.68% (SD = 0.68) ($p = 0.038$) (Table 2).

6. Discussion

Practicing mindfulness eating and yoga exercise for 8 weeks had benefits on health outcomes of pregnant women with GDM A1 (Sonja, Lucie, Jocelyne, & Evelyne, 2012). The findings support the previous studies showing that mindfulness eating is useful in reducing the quantity of consumed food resulting in reduction of fasting and postprandial blood glucose and glycosylated hemoglobin (HbA1c) (Rakhshani, Nagarathna, Mhaskar, Thomas, & Gunasheela, 2012). Yoga exercise is also helpful in reducing blood glucose because it promotes blood circulation and improves insulin sensitivity resulting in increased utilization of glucose by muscle cells (Balaji, Varne, & Ali, 2012; Kuntsevich, Bushell, & Theise, 2010). Mindfulness eating and yoga exercise promote mind–body connection resulting in physical and mental relaxation and reduction of physical and mental stress. It is known that yoga practice has a positive effect on health through the mechanism of humoral and nervous system activity (Kuntsevich et al., 2010). Therefore, it is useful in lowering blood glucose (Alexander, Taylor, Innes, Kulbok, & Selfe, 2008; Shapiro, 2009).

The largest effect of mindfulness eating and yoga exercise is that on reduction of postprandial blood glucose ($r^2 = 0.45$). Medium effects ($r^2 = 0.17, 0.20$) were observed in reduction of fasting blood glucose and glycohemoglobin (HbA1c), respectively (Table 3). HbA1c was helpful to confirm blood sugar levels during the prior 8–12 weeks because random plasma glucose was affected by intended restriction of diet before blood testing (Gilbert, 2011).

7. Implications

In order to promote diabetes self-management among pregnant women with GDM A1 practicing mindfulness eating and yoga exercise regularly may be beneficial. Longitudinal study of the health benefits

Please cite this article as: Youngwanichsetha, S., et al., The effects of mindfulness eating and yoga exercise on blood sugar levels of pregnant women with gestational diabetes mellitus, *Applied Nursing Research* (2014), http://dx.doi.org/10.1016/j.apnr.2014.02.002

ARTICLE IN PRESS

Table 3
Pre-and post-test values ($n = 170$).

Variable	Intervention group		Control group		*p*-value
	Pretest	Posttest	Pretest	Posttest	
	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	
Fasting plasma glucose (mg/dL)	88.77 (14.47)	83.39 (7.69)	89.36 (14.49)	87.85 (7.94)	0.012
2-h postprandial plasma glucose (mg/dL)	115.45 (7.58)	103.67 (9.93)	117.18 (12.14)	114.36 (10.15)	0.001
HbA1c (%)	-	5.23 (0.22)	-	5.68 (0.38)	0.016

of mindfulness eating and yoga exercise on prevention of other diabetic complications should be carried out.

8. Limitations of the study

This study had some limitations regarding the intervention. Practicing mindfulness eating and yoga exercise by themselves was prescribed in the home protocol instead of as a group intervention at the research center. Peer motivation to practice and exercise regularly was therefore lacking. However, participants reported that practicing mindfulness eating and yoga exercise by themselves at home was more convenient than doing so as a group because of their limited free time.

Further studies should compare the effect of mindfulness eating and yoga exercise between practicing alone and in a group.

9. Conclusion

Mindfulness eating and yoga exercise was effective for glycemic control among pregnant women with GDM A1. These findings contribute to the evidence for the efficacy of mindfulness eating and yoga exercise in improving glycemic control. It should be recommended in diabetes self-management.

References

Alexander, G. K., Taylor, A. G., Innes, K. E., Kulbok, P., & Selfe, T. K. (2008). Contextualizing the effects of yoga therapy on diabetes management: A review of the social determinants of physical activity. *Family & Community Health*, *31*(3), 228–239.

American Association of Diabetes (2012). Standards of medical care in diabetes-2012. *Diabetes Care*, *35*(Suppl 1), S11–S63.

Balaji, P. A., Varne, S. R., & Ali, S. S. (2012). Physiological effects of yogic practices and transcendental meditation in health and disease. *National American Journal of Medical Sciences*, *4*(10), 442–448.

Ballas, J., Moore, T. R., & Ramos, G. A. (2012). Management of diabetes in pregnancy. *Current Diabetes Reports*, *12*(1), 33–42.

Beddoe, A. E., Yang, C. P., Kenedy, H. P., Weiss, S. J., & Lee, K. A. (2009). The effects of mindfulness-based yoga during pregnancy on maternal psychological and physical distress. *Journal of Obstetric, Gynecologic, and Neonatal Nursing*, *38*(3), 310–319.

Bhake, R., & Dayan, C. M. (2010). The management of gestational diabetes mellitus after pregnancy. *Practical Diabetes International*, *27*(8), 364–367.

Chu, S. Y., Callaghan, W. M., Kim, S. Y., Schmid, C. H., & Lau, J. (2009). Position of the American Dietetic Association and American Society for Nutrition: Obesity, reproduction, and pregnancy outcomes. *Journal of the American Dietetic Association*, *109*(5), 918–927.

Daubenmier, J., Kristeller, J., Hecht, F. M., Maniner, N., Kuwata, M., & Jhaver, K. (2011). Mindfulness intervention for stress eating to reduce cortisol and abdominal fat among overweight and obese women: An exploratory randomized controlled study. *Journal of Obesity*, *10*, 1–15.

Evans, M. J. (2009). Diabetes and pregnancy: A review of pathology. *British Journal of Diabetes and Vascular Disease*, *9*, 201–212.

Gilbert, E. S. (2011). *Manual of high risk pregnancy and delivery* (4th ed.). St. Louis, MO, Mosby.

Gore, M. M., Vaze, D. R., Kulkarni, S. A., & Oak, J. P. (2008). *Yoga therapy for selected disease*. Maharashtra: Kaivalyadhama.

Jacqueminet, S., & Jannot-LaMotte, M. F. (2010). Therapeutic management of gestational diabetes. *Diabetes and Metabolism*, *36*, 658–671.

Kuntsevich, V., Bushell, W. C., & Theise, N. D. (2010). Mechanisms of yogic practices in health, aging, and disease. *Mount Sinai Journal of Medicine*, *77*(5), 559–569.

Lueprasitsakul, K., Teeyapun, K., Kittivarakul, E., Srisupundit, K., & Patumanond, J. (2008). Gestational diabetes in Lumphun hospital: Prevalence, clinical risk factors and pregnancy outcomes. *Chiang Mai Medical Journal*, *47*, 65–73.

Rakhshani, A., Nagarathna, R., Mhaskar, R., Thomas, A., & Gunasheela, S. (2012). The effects of yoga in prevention of pregnancy complications in high-risk pregnancies: A randomized controlled trial. *Preventive Medicine*, *55*, 333–340.

Sahay, G. S., & Tiwari, S. (2009). *Yoga modules for institution of higher learning*. Maharashtra: Jyeshtha Purima.

Serlin, D. C., & Lash, R. W. (2009). Diagnosis and management of gestational diabetes mellitus. *American Family Physician*, *80*(1), 57–62.

Shapiro, S. L. (2009). The integration of mindfulness and psychology. *Journal of Clinical Psychology*, *65*(5), 555–560.

Solloway, S. G., & Fisher, W. P. (2007). Mindfulness practice: A rasch variable construct innovation. *Journal of Applied Measurement*, *8*(4), 359–372.

Sonja, B. R., Lucie, M., Jocelyne, C., & Evelyne, R. (2012). Pregnancy outcomes in women with and without gestational diabetes according to the International Association of Diabetes and Pregnancy Study Groups criteria. *Obstetrics and Gynecology*, *120*(4), 746–752.

Veeraswamy, S., Vijayam, B., Gupta, V. K., & Kapur, A. (2012). Gestational diabetes: The public health relevance and approach. *Diabetes Research and Clinical Practice*, *97*, 350–358.

Webb, J. (2013). Diagnosis and treatment of gestational diabetes. *Nurse Prescribing*, *11*(1), 14–20.

Please cite this article as: Youngwanichsetha, S., et al., The effects of mindfulness eating and yoga exercise on blood sugar levels of pregnant women with gestational diabetes mellitus, *Applied Nursing Research* (2014), http://dx.doi.org/10.1016/j.apnr.2014.02.002